INCH-POUND
MIL-DTL-23053/16B
17 June 1996
SUPERSEDING
MIL-I-23053/16A
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, CROSSLINKED, ELASTOMERIC POLYOLEFIN, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

Polymer type: The base elastomer used in formulating this sleeving shall be an elastomeric polyolefin.

Continuous Operating temperature range: -75°C (-103°F) to +120°C (+248°F).

Color: The sleeving shall be furnished in a black color that conforms to Class II of MIL-STD-104 (see 1.2.2 and 3.3.l.5).

Longitudinal change: ± 10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:

M23053/16 - X XX - X

- X: Color code designator (see 1.2.1)
- XX: Size identifier (see table I)
- X: Class (When no classes are specified in the specification sheet, the numeric “0” is be used)
- M23053/16: Specification sheet

Example: Black, 1.000 inch (25.4 mm) as supplied ID sleeving shall be identified as M23053/16-005-0.

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TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness, nominal
M23053/16-001-0	.250(6.35)	.125(3.18)	.035 (.889)
M23053/16-002-0	.375(9.53)	.187(4.75)	.040 (1.02)
M23053/16-003-0	.500(12.7)	.250(6.35)	.048 (1.23)
M23053/16-004-0	.750(19.1)	.375(9.53)	.057 (1.45)
M23053/16-005-0	1.000(25.4)	.500(12.7)	.070 (1.78)
M23053/16-006-0	1.500(38.1)	.750(19.1)	.095 (2.41)
M23053/16-007-0	2.000(50.8)	1.000(25.4)	.110 (2.79)
M23053/16-008-0	3.000(76.2)	1.500(38.1)	.125 (3.18)
M23053/16-009-0	4.000(101.6)	2.000(50.8)	.140 (3.56)

1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ The color code identified is the standard acquisition color.

Unrestricted shrinkage procedures: Test method 4.6.5; 175° ± 2±C (347° ± 4°F) for 10 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test Procdeure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Low temperature flexibility	No cracks	4.6.7.1 -65° ± 2°C (-85° ± 4°F)
Heat shock	No cracks, flowing or dripping	4.6.8 200° ± 2°C (392° ± 4°F)
Restricted shrinkage	No cracks	4.6.6.1.1 150° ± 2°C (302° ± 4°F)
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	1,700(11.7)	4.6.13 ASTM D412
Ultimate elongation percent, minimum	250	4.6.13 ASTM D412
Tensile stress at 100 percent elongation, psi (MPa), maximum	1,500(10.4)	4.6.12.2 ASTM D412
Dielectric strength volts/roil (Kv/mm), minimum	300 (11.9)	4.6.2 ASTM D2671
Volume resistivity, Ohm-cm, minimum	1 x 10^{9}	4.6.2 ASTM D876
Specific gravity, maximum	1.5	4.6.2 ASTM D792
Corrosion	No corrosion or pitting	4.6.10.1 121° ± 2°C (250° ± 4°F); 4.6.10.2 135° ± 2°C (275° ± 4°F); for 16 hours
Water absorption, percent, maximum	2.0	4.6.2 ASTM D570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.9 150° ± 2°C (302° ± 4°F), 168 hours
Tensile strength, psi (MPa), minimum	1,500(10.3)	
Ultimate elongation, percent, minimum	200	

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Fluid resistance, properties after:		4.6.11 2/
Tensile strength, psi (MPa), minimum	1,500(10.4)	
Ultimate elongation, percent, minimum	200	
Dielectric strength, volts/mil (Kv/mm), minimum	200(7.9)	
Flammability	Self-extinguishing within one minute and no more than 25 percent of indicator flag burned or charred	4.6.14 Procedure B ASTM 2671
Fungus resistance	Rating less than 1	4.6.2 ASTM G21

1/ Unless otherwise specified, requirements, test procedures and conditions are for all classes.

2/ In addition to the fluids specified in 4.6.11, the sleeving shall also be immersed in diesel fuel conforming to MIL-F-46162.

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 2 years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: Heat shrinkable flexible elastomeric sleeving is intended for use on heavy duty cables or harness systems, such as ground support which see high levels of physical abuse and exposure to fuels and oils, coupled with high and low temperature extremes.

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Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - ME, MI, MU
Navy - EC, MC
Air Force - 80
DLA - GS

Preparing activity:
Navy - AS

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